



FB-ZWUNet: A deep learning network for corpus callosum segmentation in fetal brain ultrasound images for prenatal diagnostics

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ARTICLE INFO

Keywords:

Fetal Brain Ultrasound
Corpus Callosum Segmentation
Prenatal Diagnostics

ABSTRACT

Objective: Develop an automated method to replace manual segmentation of the corpus callosum (CC) in fetal brain ultrasound images, thereby reducing reliance on specialist expertise and mitigating human error, ultimately improving CC segmentation accuracy and enhancing fetal brain development assessments.

Methods: We propose FB-ZWUNet, an end-to-end CC segmentation network composed of three key modules: (1) Zernike Attention Module (ZAM) for feature enhancement, (2) Wavelet Attention Module (WAM) for optimized feature fusion and precise image reconstruction, and (3) Morphological Constraint Module (MCM) for accurate edge and region capture. The network was trained on the FB-CC dataset, which consists of 1,336 annotated fetal brain mid-sagittal ultrasound images.

Results: FB-ZWUNet achieved a Dice coefficient of 0.8743 and IoU of 0.7813, outperforming current state-of-the-art methods in accuracy and stability while offering faster inference times. The FB-CC dataset provides comprehensive samples from 18 to 32 weeks of gestation, collected using three different ultrasound devices.

Conclusions: FB-ZWUNet effectively addresses segmentation challenges, demonstrating significant clinical potential in assessing fetal neurological development and improving the reliability of early diagnosis and prenatal care.

1. Introduction

In clinical prenatal ultrasound diagnostics, accurately assessing the development of key fetal brain structures is critical for evaluating neurodevelopment and detecting potential anomalies or congenital conditions. As shown in Fig. 1(a), the corpus callosum (CC)—the largest white matter structure connecting the left and right hemispheres—is a primary target in fetal brain examinations [1]. Between 18 and 32 weeks of gestation, the CC undergoes significant maturation, making this period essential for evaluating its developmental status [2]. Accurate assessment during this stage is crucial for diagnosing conditions such as Agenesis of the Corpus Callosum (ACC) and Dysgenesis of the Corpus Callosum, both of which are often associated with neurological and cognitive impairments. Early detection of these abnormalities is vital for clinical management and parental counseling [3].

During the diagnostic process, clinicians use advanced ultrasound equipment to acquire three-dimensional (3D) volumetric data of the

fetal brain. From these data, the fetal brain mid-sagittal (FBMS) plane is extracted, providing the clearest view of the CC and related structures [4,5]. Once the FBMS plane is identified, the CC region is manually annotated and segmented to derive critical biological measurements such as area, perimeter, and length/width dimensions. These parameters are key indicators for evaluating the developmental state and identifying potential abnormalities.

As illustrated in Fig. 1(b), the CC region in FBMS ultrasound images can vary substantially in color, texture, and morphology due to differences in equipment, patient characteristics, and examination timing. Furthermore, traditional detection processes heavily depend on the clinician's experience and subjective judgment, leading to potential misdiagnoses and inefficiencies [6]. Inaccurate assessments of CC development can severely impact the evaluation of fetal brain structures and neurodevelopment, potentially causing irreversible effects on neonatal health.

To overcome these challenges, we developed FB-ZWUNet—an end-

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<https://doi.org/10.1016/j.bspc.2025.107499>

Received 19 August 2024; Received in revised form 17 December 2024; Accepted 2 January 2025

Available online 19 January 2025

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to-end Fetal Brain Zernike Wavelet UNet—integrated into a mobile computer-aided diagnosis (CAD) system. This network is designed to fully leverage the morphological features and clinical priors of CC, ensuring more accurate segmentation. FB-ZWUNet consists of three key modules: the Zernike Attention Module (ZAM) for highlighting essential features, the Wavelet Attention Module (WAM) for optimizing feature fusion and mitigating noise, and the Morphological Constraint Module (MCM) for capturing precise edge and region features. Additionally, we curated the Fetal Brain Corpus Callosum Dataset (FB-CC), comprising 1,336 meticulously annotated 2D FBMS ultrasound images, to provide a robust foundation for training and validating our model, thereby enhancing its performance and reliability.

1.1. Related Works

1.1.1. Medical image segmentation

Recent advancements in medical image segmentation have been significantly driven by deep learning technologies. The introduction of the U-Net model [7] marked a turning point, inspiring numerous innovations. The UNet 3+ model [8] extended its capabilities, enhancing both accuracy and computational efficiency. Transformer-based architectures, such as TransUNet [9] and Swin-UNet [10], brought advanced processing capabilities, enabling superior segmentation of complex image datasets and ushering in a new era of segmentation technology. Specific applications, such as multi-instance instrument segmentation [11] and real-time polyp detection [12], further demonstrate the versatility of deep learning in niche medical imaging domains. Innovations like HyperDense-Net [13] and multi-scale self-guided attention networks [14] have significantly improved segmentation in multimodal imaging, emphasizing the efficacy of deep learning in precise medical image analysis. More recent network designs, such as DCsAU-Net [15] and nnFormer [16], illustrate a shift from foundational models to highly sophisticated systems, achieving greater segmentation accuracy and computational efficiency. However, these methods often lack specificity for certain imaging modalities, such as ultrasound, where the advantages of such architectures are less evident. This underscores the need for targeted approaches tailored to ultrasound imaging.

1.1.2. Ultrasound image segmentation

Ultrasound image segmentation has transitioned from traditional

methods to deep learning-based approaches, significantly improving performance. Early techniques relied on handcrafted imaging features and mathematical models [17]. The adoption of deep learning marked a paradigm shift, enabling more robust and flexible processing. For example, Hwang and Kim [18] introduced weakly supervised methods, paving the way for more advanced segmentation approaches. Further improvements followed with optimized deep learning models, such as Khaledyan et al.'s [19] enhancements to U-Net. Specialized applications, like Nishi et al.'s [20] segmentation of intravascular ultrasound (IVUS) images using convolutional neural networks (CNNs), demonstrated the potential of deep learning in niche contexts. Advanced models like PBNNet [21], designed for breast tumor segmentation, integrate multi-level global perception and boundary guidance modules, improving the detection of challenging cases like non-enhanced tumors. Similarly, the SAM model [22] highlights ongoing progress, applying novel segmentation strategies to ultrasound imaging. Despite these advancements, the segmentation of complex and variable fetal ultrasound images remains challenging due to the inherent variability in fetal anatomy and imaging conditions. This necessitates further exploration of methods that can better utilize image and clinical information to optimize segmentation performance.

1.1.3. Fetal ultrasound image segmentation

Fetal ultrasound segmentation has evolved significantly, transitioning from basic feature-based methods to leveraging advanced deep learning technologies. Early efforts, such as Yu et al.'s [23] computational methods for fetal weight estimation, laid the groundwork for further advancements. Comprehensive evaluations by Rueda et al. [24] established benchmarks for segmentation methodologies, while cascading fully convolutional networks employed by Wu et al. [25] introduced robust approaches for processing complex fetal ultrasound images. Multi-task deep learning approaches, such as those by Sobhaninia et al. [26], have been used to estimate biometric parameters, improving segmentation accuracy and robustness. Other innovations include Zeng et al.'s [27] depth-supervised, attention-gated V-Net and Ashkani Chenarlogh et al.'s [28] refined U-Net models. Lightweight convolutional architectures, like Zeng et al.'s [29] automated method for fetal head circumference measurement, demonstrate the feasibility of applying efficient deep learning models to real-world clinical tasks. Zimmer et al. [30] addressed challenges in placental imaging, highlighting solutions for limited view issues and segmentation uncertainty.

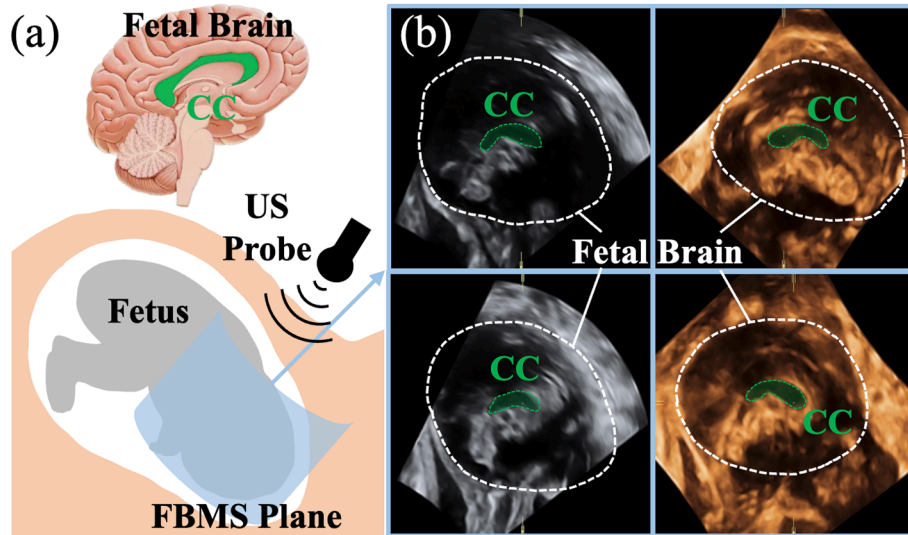


Fig. 1. Schematic Diagram of Fetal Brain Ultrasound. Ultrasound imaging of the fetal brain and two-dimensional mid-sagittal plane diagrams. (a): the schematic of the sagittal plane structure of the fetal brain, including the relative positioning of the ultrasound probe to the mid-sagittal plane (FBMS Plane). (b): reconstructed two-dimensional FBMS planes from 4 different fetuses.

Despite these advancements, challenges remain in segmenting key fetal brain structures due to inter-individual variability, developmental differences, and imaging inconsistencies. Integrating clinical knowledge and advanced image processing techniques remains critical for overcoming these limitations.

1.1.4. Fetal brain ultrasound image segmentation

The application of deep learning in fetal brain ultrasound segmentation has evolved from basic model adaptations to more sophisticated feature extraction techniques. Shu et al. [31] introduced the Efficient Channel Attention U-Net (ECAU-Net), which leverages channel attention mechanisms to improve segmentation accuracy in the cerebellar region, emphasizing the importance of attention mechanisms in recognizing key anatomical features. Wang et al. [32] refined U-Net to optimize fetal head image segmentation, showcasing the benefits of task-specific network modifications. Our previous work [33] highlighted the integration of traditional imaging methods with deep learning for the automated localization and segmentation of key fetal brain structures, including the cavum septum pellucidum and cerebellar vermis. This research demonstrated the value of combining morphological features with relative positioning in preprocessing. Further advancements include Qiao et al.'s [34] development of a dilated squeeze-excitation U-Net, which enhances feature extraction and segmentation of complex image data, and Gofer et al.'s [35] machine learning-based approach for early fetal brain classification, demonstrating the potential for early-stage anomaly detection. Recently, FetSAM [36] achieved enhanced segmentation accuracy through prompt-based learning and dual loss mechanisms, setting a new benchmark in fetal head biometric analysis. Additional contributions have further refined automated segmentation techniques, incorporating anatomical priors and supervised learning strategies for fetal head and femur segmentation [37–40]. While these studies have contributed significantly to the field, challenges remain in effectively handling limited and variable data, as well as in incorporating the detailed morphological characteristics and anatomical priors unique to the fetal brain. Addressing these issues is crucial for achieving more accurate and reliable segmentation results.

2. Material and methods

2.1. Overview

Fig. 2 presents an overview of our proposed Fetal Brain Zernike Wavelet UNet (FB-ZWUNet). The network comprises three main parts: (1) an encoder integrated with the Zernike Attention Module (ZAM) shown in the red area; (2) a decoder combined with the Wavelet Attention Module (WAM) shown in the blue area; (3) an output layer with the Morphological Constraint Module (MCM) shown in the orange area. In FB-ZWUNet, the input image I is processed by the encoder, extracting key features through convolutional layers and max-pooling, with enhanced feature capture via the ZAM. The decoder performs up-sampling and convolution, utilizing the WAM for feature fusion and image reconstruction, ensuring detailed and high-quality images. Finally, the MCM uses morphological operations to enhance edge and regional feature representation, ensuring structural integrity and accuracy in segmentation results. The specific operations and the corresponding feature map sizes are detailed in Table 1.

Specifically, the ZAM module focuses on key feature areas, improving extraction in complex backgrounds and ambiguous regions. The WAM module optimizes feature fusion through multi-scale analysis, enhancing image reconstruction quality. The MCM module improves edge perception and regional feature accuracy through morphological operations, ensuring consistency and stability in segmentation results. This multi-module collaboration allows FB-ZWUNet to excel in fetal brain ultrasound image segmentation, leveraging the morphological and texture characteristics of the CC, thus improving segmentation accuracy and diagnostic reliability.

2.2. Encoder with Zernike Attention Module

In FBMS ultrasound images, the boundaries of the CC are often indistinct and have low contrast relative to surrounding regions. Additionally, inherent ultrasound noise makes accurate segmentation more challenging. To address these issues, we introduce the Zernike Attention

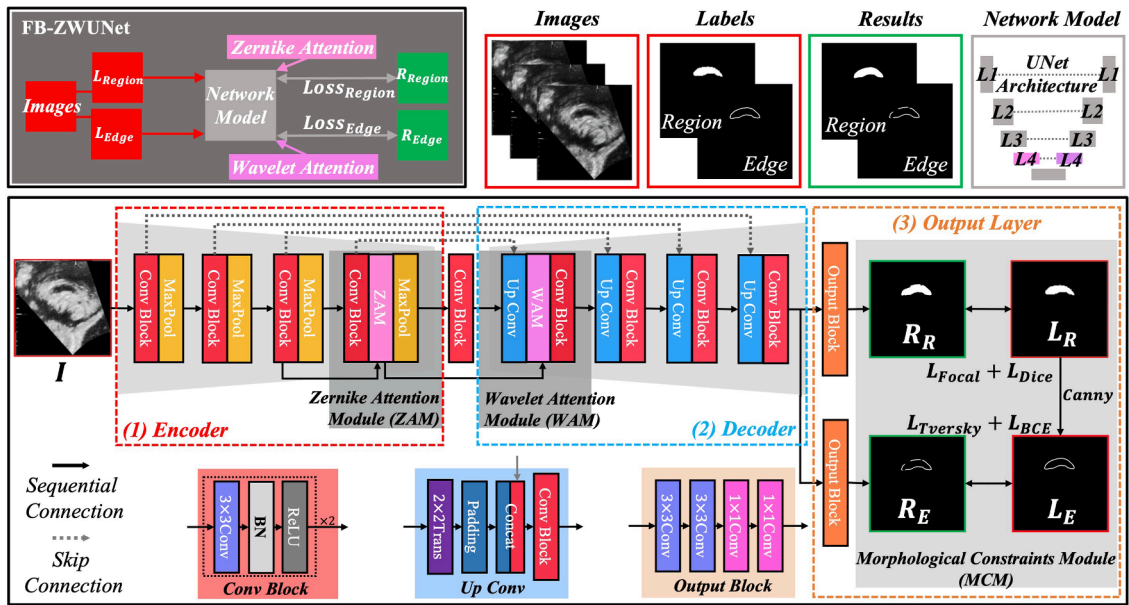


Fig. 2. Overview of the FB-ZWUNet Architecture. The FB-ZWUNet is based on the UNet architecture and achieves Corpus Callosum segmentation through the innovation of three key modules: Manuscript with Tracked Changes (1) an Encoder with the Zernike Attention Module (ZAM) (red dashed box), (2) a Decoder with the Wavelet Attention Module (WAM) (blue dashed box), and (3) an Output Layer incorporating the Morphological Constraint Module (MCM) (orange dashed box). The network input, shown in red, includes the original images (I) with region labels (L_R) and edge labels (L_E). The network prediction output, shown in green, includes the prediction of regions (R_R) and edges (R_E). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Feature Map Sizes and Operations for FB-ZWUNet.

Stage	Operation	Output Size ($H \times W \times C$)
Input	—	$128 \times 128 \times 1$
Encoder L1	Conv Block + MaxPool	$64 \times 64 \times 64$
Encoder L2	Conv Block + MaxPool	$32 \times 32 \times 128$
Encoder L3	Conv Block + MaxPool	$16 \times 16 \times 256$
Encoder L4	Conv Block + ZAM + MaxPool	$8 \times 8 \times 512$
Bottleneck	Conv Block	$8 \times 8 \times 512$
Decoder L4	UpConv + WAM + Conv Block	$16 \times 16 \times 256$
Decoder L3	UpConv + Conv Block	$32 \times 32 \times 128$
Decoder L2	UpConv + Conv Block	$64 \times 64 \times 64$
Decoder L1	UpConv + Conv Block	$128 \times 128 \times 64$
Output Layer	Output Block	$128 \times 128 \times 1$

Conv Block: [Convolution(3×3) + Batch Normalization + ReLU] $\times 2$.

MaxPool: MaxPooling(2×2 , stride = 2).

UpConv: Transposed Convolution (2×2 , stride = 2) + Padding + Concatenate (Skip Connection) + Conv Block.

Output Block: Convolution(3×3) $\times 2$ + Convolution(1×1) $\times 2$.

Module (ZAM), which leverages Zernike polynomials to enhance key feature extraction and improve structural contrast, thereby reducing segmentation errors and increasing accuracy.

Zernike polynomials are a set of orthogonal polynomials defined on the unit disk and represented in polar coordinates (r, θ) . In our approach, we assume the image center $(N/2, N/2)$ as the origin of the polar coordinate system. For a pixel (i, j) :

$$r(i, j) = \sqrt{\left(\frac{i - N/2}{N/2}\right)^2 + \left(\frac{j - N/2}{N/2}\right)^2}, \theta(i, j) = \arctan2(j - N/2, i - N/2) \quad (1)$$

Here, $r(i, j)$ is the normalized radius, and $\theta(i, j)$ is the angular coordinate relative to the image center. Let $x(i, j)$ represent the intensity at position (i, j) . The Zernike moment $Z_{p,q}$ of order p, q is:

$$Z_{p,q} = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} x(i, j) \cdot \text{Re}(e^{-j \cdot q \cdot \theta(i, j)}) \quad (2)$$

In Eq. (2), p, q are the orders of the Zernike polynomials, and $\text{Re}(\cdot)$ denotes the real part operation [41,42].

By meticulously analyzing and weighting image regions, ZAM focuses on the shape and texture characteristics of the CC, enhancing its distinction from surrounding tissues. Zernike moments are known to be robust descriptors of shape and have been widely applied in medical image analysis due to their rotational invariance and strong representational capability for complex boundaries [43]. In this way, ZAM provides a richer feature map for the decoder, significantly improving segmentation accuracy.

Fig. 2 (red area) illustrates the encoder with ZAM. The encoder consists of four convolutional blocks, each with two 3×3 convolutions followed by ReLU activation, and a 2×2 MaxPooling layer for down-sampling (except for the last block). ZAM is placed at the end of the encoder because higher-level convolutional layers extract more

advanced semantic features. By applying Zernike moments at this stage, ZAM can highlight key CC regions based on shape and texture cues. This placement also reduces network complexity, improves computational efficiency, and integrates smoothly with subsequent modules.

Fig. 3 and Algorithm 1 illustrate the workflow of ZAM. The global feature map $g(C_g, W, H)$ is obtained by applying global average pooling to the encoder output, capturing overall context. The local feature map $l(C_l, W, H)$ is the direct encoder output at that layer, preserving spatial details. We apply separate 1×1 convolutions to both g and l to project them into a common intermediate channel space F_{int} : $g'(F_{int}, W, H)$ and $l'(F_{int}, W, H)$. By aligning channels to F_{int} , we can effectively fuse global and local information. Adding g' and l' gives the fused feature map $f(F_{int}, W, H)$. After a ReLU activation, another 1×1 convolution followed by a Sigmoid function transforms f into a single-channel attention map $f''(1, W, H)$. We then compute Zernike moments on f'' to produce $f_z''(1, W, H)$, a shape-focused map that encodes CC structural information. Finally, we multiply f_z'' element-wise with $l(C_l, W, H)$. The single-channel f_z'' is broadcast across the C_l channels of l , re-weighting features to emphasize CC structures and suppress noise:

Algorithm 1 Zernike Attention Module (ZAM)

Input:

Global Feature Map $g(C_g, W, H)$

Local Feature Map $l(C_l, W, H)$

Steps:

1 : Convolve g by 1×1 Conv $\rightarrow g'(F_{int}, W, H)$

2 : Convolve l by 1×1 Conv $\rightarrow l'(F_{int}, W, H)$

3 : Add g' and $l' \rightarrow f(F_{int}, W, H)$

4 : ReLU Activation $\rightarrow f'(F_{int}, W, H)$

5 : 1×1 Conv and Sigmoid $\rightarrow f''(1, W, H)$

6 : Compute Zernike Moment $\rightarrow f_z''(1, W, H)$

7 : Multiply f_z'' with $l \rightarrow Z(C_l, W, H)$

Output:

Zernike Attention Feature Map $Z(C_l, W, H)$

$$Z(C_l, W, H) = f_z''(1, W, H) \odot l(C_l, W, H) \quad (3)$$

By integrating ZAM into the encoder, our network leverages shape-aware features derived from Zernike moments, improving the overall segmentation accuracy of CC in FBMS ultrasound images.

2.3. Decoder with Wavelet Attention Module

Effective ultrasound image segmentation not only depends on precise feature extraction by the encoder but also relies on robust feature reconstruction by the decoder. To address the challenges of handling noise, artifacts, and subtle structural details, we introduce the Wavelet Attention Module (WAM) into the decoder. WAM leverages multi-scale feature analysis inspired by wavelet transforms to enhance feature fusion, improve robustness, and ultimately boost segmentation accuracy for the CC.

Unlike traditional wavelet transforms, we simulate wavelet-like

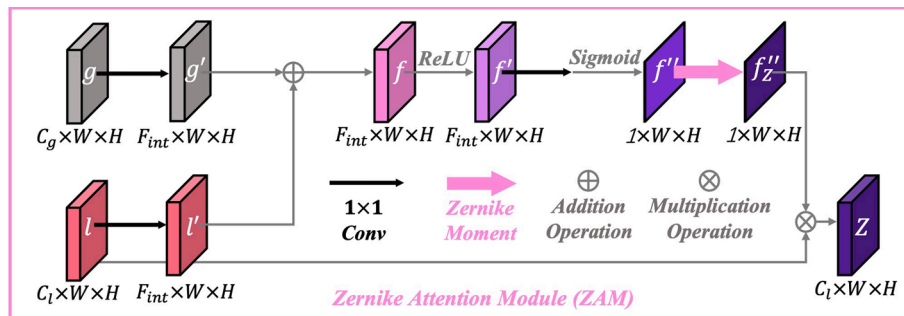


Fig. 3. Details of the Zernike Attention Module (ZAM).

multi-scale analysis using multiple convolution kernels of different sizes (3×3 , 5×5 , and 7×7) [44,45]. These operations capture features at various spatial frequencies, akin to decomposing the image into multiple resolution levels, thus enriching the final representation without explicitly performing a wavelet transform [46].

Fig. 2 (blue area) shows the decoder integrated with WAM. The decoder reconstructs image features through four up-sampling blocks, each comprising a transposed convolution for up-sampling followed by convolutional layers. We integrate WAM between the first up-sampling block and its subsequent convolutional block. Placing WAM at this early stage in the decoder allows it to access higher-level semantic features received from the encoder, promoting effective interaction between encoder-derived context and decoder reconstruction, and ensuring that multi-scale information is leveraged from the outset.

Fig. 4 and Algorithm 2 illustrate the WAM workflow. Similar to the encoder stage (ZAM), we start by obtaining a global feature map $g(C_g, W, H)$ via global average pooling, capturing broad contextual cues, and a local feature map $l(C_l, W, H)$ directly from the decoder's current layer, preserving spatial details. We apply 1×1 convolutions to both g and l to align their channels to an intermediate dimension F_{int} , producing $g'(F_{int}, W, H)$ and $l'(F_{int}, W, H)$. Adding these results yields the fused feature map $f(F_{int}, W, H)$. After applying a ReLU activation, another 1×1 convolution, and a Sigmoid function, we obtain a single-channel attention map $f''(1, W, H)$. We then “expand” this single-channel map back into $f_E''(F_{int}, W, H)$ to facilitate multi-scale analysis. We pass f_E'' through three parallel convolution layers with kernels of 3×3 , 5×5 , and 7×7 , resulting in $f_{3 \times 3}''(F_{int}, W, H)$, $f_{5 \times 5}''(F_{int}, W, H)$, and $f_{7 \times 7}''(F_{int}, W, H)$. These three feature maps represent a decomposition of features at multiple spatial scales, analogous to a wavelet's multi-frequency decomposition. By adding these three multi-scale feature maps, we obtain $f_W''(1, W, H)$. Finally, we perform an element-wise multiplication between f_W'' and the local feature map $l(C_l, W, H)$. The single-channel attention applied via f_W'' is broadcast across the C_l channels of l , effectively integrating multi-scale contextual cues into the local feature representation. This process emphasizes structures of interest at various scales, improving both the detail preservation and noise suppression in the reconstructed CC segmentation.

Algorithm 2 Wavelet Attention Module (WAM)

Input:

Global Feature Map $g(C_g, W, H)$

Local Feature Map $l(C_l, W, H)$

Steps:

- 1 : Convolve g by 1×1 Conv $\rightarrow g'(F_{int}, W, H)$
- 2 : Convolve l by 1×1 Conv $\rightarrow l'(F_{int}, W, H)$
- 3 : Add g' and $l' \rightarrow f(F_{int}, W, H)$
- 4 : ReLU Activation $\rightarrow f(F_{int}, W, H)$
- 5 : 1×1 Conv and Sigmoid $\rightarrow f''(1, W, H)$
- 6 : Expand $\rightarrow f_E''(F_{int}, W, H)$
- 7 : Convolve f_E'' by 3×3 , 5×5 , 7×7 Conv $\rightarrow f_{3 \times 3}''(F_{int}, W, H)$, $f_{5 \times 5}''(F_{int}, W, H)$, $f_{7 \times 7}''(F_{int}, W, H)$
- 8 : Add $f_{3 \times 3}''$, $f_{5 \times 5}''$ and $f_{7 \times 7}'' \rightarrow f_W''(F_{int}, W, H)$
- 9 : Multiply f_W'' with $l \rightarrow W(C_l, W, H)$

Output:

Wavelet Attention Feature Map $W(C_l, W, H)$

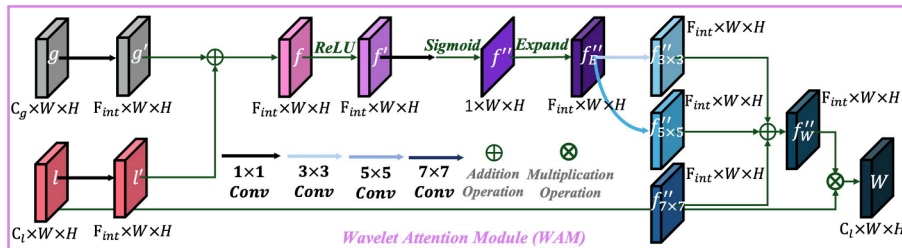


Fig. 4. Details of the Wavelet Attention Module (WAM).

Integrating WAM in the decoder thus enriches the segmentation process with multi-scale information, complementing the shape-aware enhancements introduced by ZAM in the encoder. Together, these modules provide a robust framework that leverages both shape priors and multi-scale feature analysis, leading to more accurate and stable CC segmentation in challenging FBMS ultrasound images.

2.4. Output layer with morphological constraints Module

In our method, the Morphological Constraints Module (MCM) is introduced to address the diverse shapes and sizes of CC. MCM supervises the prediction of edge-aware mask models in both branches, using edge-aware geometric constraints. Utilizing the Canny operator [47,48], binary edge maps of the CC are generated. MCM employs a compound boundary loss function (as shown in Fig. 2, MCM), combining binary cross-entropy loss and Dice loss, to enhance edge prediction. Fig. 2 (orange area) illustrates the output layer with MCM.

In FB-ZWUNet, the overall loss formula is:

Region Loss combines Focal Loss and Dice Loss:

$$L_{Region} = \alpha_1 L_{Focal}(R_R, L_R) + \beta_{1Dice}(R_R, L_R) \quad (5)$$

Edge Loss integrates Tversky Loss and BCE Loss:

$$L_{Edge} = \alpha_2 L_{Tversky}(R_E, L_E) + \beta_{2BCE}(R_E, L_E) \quad (6)$$

Here, R_R and R_E are the predictions for the region and edge, respectively, and L_R and L_E are their corresponding ground truths. α and β are hyperparameters to balance the weights of the loss components. This approach ensures accurate capture of the morphology and boundary details of the CC in segmentation.

Focal Loss reduces the impact of easy-to-classify samples, enabling the model to focus on challenging regions of the CC, essential for managing class imbalances common in brain imaging. This characteristic of Focal Loss is pivotal for enhancing segmentation accuracy in brain scans, where distinctions between different regions can be subtle. Focal Loss is formulated as follows:

$$L_{Focal}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (7)$$

where p_t represents the probability of the prediction of model being the true category, α is a balancing factor, and γ is an adjustment parameter.

Dice Loss improves the shape matching between predictions and actual masks, ensuring the correct capture of the CC continuity and overall structure. Dice Loss is formulated as follows:

$$L_{Dice} = 1 - \frac{2 \sum_{i=1}^{H \times W} p_i g_i}{\sum_{i=1}^{H \times W} p_i + \sum_{i=1}^{H \times W} g_i} \quad (8)$$

where p_i and g_i represent the predicted and actual label values at pixel i .

Tversky Loss further optimizes the recognition of the CC areas with less distinct edges, particularly useful in cases of variable image quality or blurred boundaries. Tversky Loss is formulated as follows:

$$L_{Tversky} = 1 - \frac{\sum_{i=1}^{H \times W} p_i g_i}{\sum_{i=1}^{H \times W} p_i g_i + \alpha \sum_{i=1}^{H \times W} p_i (1 - g_i) + \beta \sum_{i=1}^{H \times W} (1 - p_i) g_i} \quad (9)$$

BCE Loss offers pixel-level classification accuracy, which is particularly important for precisely delineating the boundaries between the CC and surrounding tissues. BCE Loss is formulated as follows:

$$L_{BCE} = -\frac{1}{H \times W} \sum_{i=1}^{H \times W} [g_i \log(p_i) + (1 - g_i) \log(1 - p_i)] \quad (10)$$

where H and W represent the height and width of the image, while α and β in Tversky Loss are weights used to balance false positives and false negatives.

MCM finely models edges and shapes of CC, enhancing segmentation accuracy. It bolsters the capability to recognize edges of CC, especially in cases with complex shapes or small sizes. MCM, providing precise edge information through refined boundary maps, assists in capturing the complete form of CC, crucial for improving segmentation performance, particularly in handling small or morphologically varied CC structures.

2.5. Dataset

To advance prenatal ultrasound detection and fetal brain ultrasound image analysis, we developed the Fetal Brain Ultrasound Corpus Callosum Dataset (FB-CC), a high-quality, open-source dataset with meticulous annotations. Similar large-scale annotated datasets have proven highly valuable for improving segmentation accuracy and fostering algorithmic advancements [49]. This dataset aims to establish innovative standards and enhance the precision of automated recognition and analysis of the CC. Our objective with this groundbreaking dataset is to significantly improve the early detection and diagnostic accuracy of fetal brain diseases, thereby bringing revolutionary improvements to infant health management and treatment strategies.

Dataset Collection and Annotation: Our dataset, as shown in Fig. 5 (a), collected by professional doctors at Shengjing Hospital of China Medical University (Shenyang), primarily comprises two-dimensional fetal brain ultrasound images, with a specific focus on the detailed analysis and annotation of the CC. The detailed dataset composition, including the distribution across gestational weeks and ultrasound devices, is presented in Table 2. These images were gathered from 1,336 healthy singleton pregnancy volunteers, covering gestational weeks 18–32. The participants predominantly originated from Shenyang and surrounding regions in Liaoning Province, China. To ensure diversity and robustness, the data collection included three advanced ultrasound systems: GE Voluson E8, GE Voluson E10, and Samsung WS80A, all equipped with transabdominal three-dimensional transducers such as

Table 2
Dataset Composition.

Gestational Age (week)		Ultrasound Devices		
		GE Voluson E8	GE Voluson E10	Samsung WS80A
18–20	(244)	89	97	58
21–23	(270)	97	108	65
24–26	(272)	102	109	61
27–29	(294)	106	112	76
30–32	(256)	92	103	61

RAB 4–8, RM 6C, or eM6C [50]. The use of multiple ultrasound devices helped capture variations in imaging conditions and quality, reflecting real-world diversity in clinical ultrasound systems.

As shown in Fig. 5(b), after the acquisition of three-dimensional ultrasound image data, professional software was employed for reconstruction, yielding detailed three-dimensional volumetric data of the fetal brain. We paid particular attention to the mid-sagittal plane of the fetal brain, a crucial aspect in clinical practice for accurately displaying the development of important tissues and organs of the fetal brain, especially the CC. Five ultrasound experts, each with over ten years of experience, manually annotated these two-dimensional images, ensuring the selected images exhibited a complete fetal brain without significant sound shadow or measurement caliper coverage, thus guaranteeing the high quality and accuracy of the images and data.

Dataset Statistics: As illustrated in Fig. 5(a), considering the differences in color, texture, and other characteristic features among various ultrasound devices, we performed necessary preprocessing on all images, including gray value normalization and size normalization, to ensure data consistency and standardization. After processing, the original dataset included 1336 images resampled to 128×128 pixels, with data from GE Voluson E8 comprising 486 individuals, GE Voluson E10 comprising 529 individuals, and Samsung WS80A comprising 321 individuals. These images were randomly allocated to the training set, test set, and validation set in an 8:1:1 ratio, comprising 1070, 133, and 133 images respectively.

3. Results

3.1. Implementation details

In our study, the training phase employed the data division method of the proposed dataset, utilizing images with the original dimensions of 128×128 . The model was implemented in the PyTorch 1.10.2 envi-

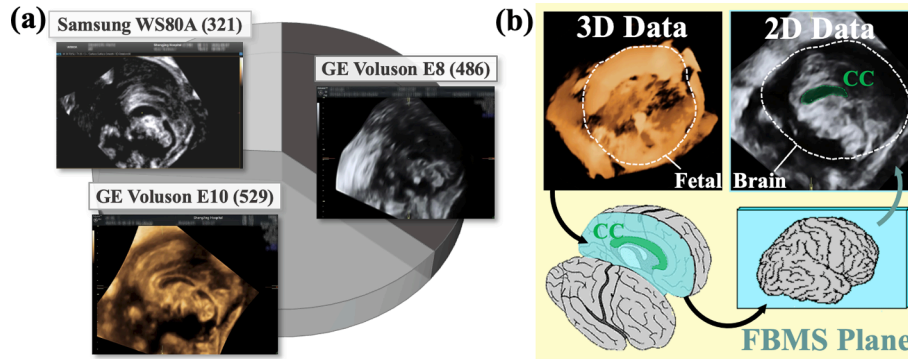


Fig. 5. Schematic Diagram of the Dataset. (a) Data Composition: The FB-CC dataset consists of 1,336 annotated mid-sagittal fetal brain ultrasound images obtained between 18 and 32 weeks of gestation. Data were collected using three different ultrasound devices—GE Voluson E8 (486 images), GE Voluson E10 (529 images), and Samsung WS80A (321 images)—to ensure diverse imaging conditions and enhance the dataset's generalizability. (b) Data Acquisition Illustration: A 3D ultrasound scan of the fetal brain is performed, from which the clinically significant mid-sagittal plane (blue plane) is extracted as a 2D image. These mid-sagittal images are then meticulously annotated by experts, forming the core of the FB-CC dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ronment and ran on an Ubuntu workstation equipped with a 10-core 3.30 GHz processor and two GPU GeForce GTX 1080 Ti, accelerated by CUDA 11.4. The training configuration included 36 workers, with a batch size set to 16, and a total of 100 training epochs. The initial learning rate was set at 0.001. The Adam optimizer [51] was used for model training, with momentum parameters β_1 and β_2 set to 0.9 and 0.99, respectively. The adjustment of the learning rate was controlled by the 'ReduceLROnPlateau' scheduler [52], based on the training loss or the Dice value of the validation set, with a reduction factor of 0.5 and a patience of 10 epochs. To evaluate the performance of FB-ZWUNet in segmenting CC in fetal brain ultrasound images, we use metrics such as Dice coefficient (Dice), Intersection over Union (IoU), Precision (PRE), Recall (REC), Specificity (SPE), Accuracy (ACC), and Hausdorff distance (HD), as detailed in Table 3.

3.2. Contrast Experiment

3.2.1. Segmentation Validation on FB-CC dataset

In our study, FB-ZWUNet demonstrates outstanding performance in CC segmentation on the FB-CC dataset, surpassing advanced methods such as UNet [7] and other state-of-the-art methods: TransUNet[9], Swin-UNet[10], ResUNet[53], Attention UNet[54], R2 UNet[55], Channel UNet[56], KiUNet[57], MedT[58], MISSFormer[59], SAM[60]. To ensure fairness in comparison, all methods used consistent training parameters and data augmentation techniques, including initial learning rate, batch size, learning rate delay method, and data augmentation methods like horizontal and vertical flips.

Data Validation: As shown in Table 4, FB-ZWUNet improves the Dice coefficient (DSC) by 3.65 percentage points over UNet, reaching 87.43 %, and increases Intersection over Union (IoU) by 4.54 points, reaching 78.13 %. Compared to the high-performing MISSFormer, FB-ZWUNet leads with 1.16 percentage points higher DSC and 2.11 points higher IoU, demonstrating superior segmentation accuracy. The network achieves a precision of 0.8738, recall of 0.8875, specificity of 0.9988, and overall accuracy of 0.9976, indicating its ability to accurately identify target structures with high specificity. The Hausdorff distance is reduced to 8.2899, indicating better boundary localization.

Efficiency Validation: As shown in the last column of Table 4, the inference time of FB-ZWUNet is only 125.72 ms (ms), like the simplest UNet and twice as fast as more complex methods like MedT and SAM. During training, our proposed modules (ZAM, WAM, and MCM) enhanced focus on CC regions and contours, improving segmentation performance without adding excessive computations during inference, thus increasing computational efficiency.

Visual Validation: Fig. 6 compares segmentation results of four samples. Columns m and n show the results of FB-ZWUNet and ground truth, respectively, while columns b to l display results from other methods. Most methods perform well on simple samples, but in challenging cases with complex CC targets, indistinct boundaries, or noise, many methods show significant errors. As indicated by the yellow and orange arrows, in some samples with indistinct foreground-background

distinction, segmentation results may have boundary overflow or indentation issues, and even erroneous segmentation as shown by the blue circles. In contrast, FB-ZWUNet provides clearer outlines and more precise details, highlighting its potential for clinical applications in fetal brain structure analysis.

3.2.1.1. Application Validation in computer aided diagnosis. Fig. 7(a) compares the accuracy and mean squared error of FB-ZWUNet with those of five ultrasonography doctors from Shengjing Hospital of China Medical University in predicting the corpus callosum (CC) region. While highly skilled doctors can achieve accuracies close to 87 % in individual cases, their performance varies due to differences in experience and inherent challenges of manual annotation, as reflected in higher mean squared errors and wider data distributions. In contrast, FB-ZWUNet achieves a mean Dice Similarity Coefficient (DSC) of 87.43 % and a mean squared error of 4.37 %, demonstrating stability and precision compared to manual methods.

Fig. 7(b) further highlights the segmentation performance of FB-ZWUNet across gestational age ranges (18–20, 21–23, 24–26, 27–29, and 30–32 weeks, as grouped in Table 2). The results showcase the reliability of our model, with closely aligned mean (red triangles) and median (red lines) values across all age ranges. The mid-gestation ranges (21–29 weeks) exhibit the best performance, with mean DSC values exceeding 87.3 % and narrower interquartile ranges (IQR), reflecting optimal image quality and well-defined structural features during these weeks. In contrast, early gestation (18–20 weeks) faces challenges due to underdeveloped structures (mean DSC: 84.32 %), while late gestation (30–32 weeks) encounters issues such as acoustic shadowing and cranial ossification, resulting in slightly more dispersed data (mean DSC: 84.76 %).

Together, these experiments underline the clinical reliability and stability of FB-ZWUNet in segmenting the CC region, outperforming manual segmentation in accuracy and consistency, and maintaining robust performance across different gestational stages. These results demonstrate the potential of FB-ZWUNet for enhancing prenatal diagnostics and clinical workflows.

3.3. Ablation Experiment

3.3.1. Impact of each modules

Table 5 and Fig. 8 compare the performance of different configurations on the FB-CC dataset, showing how the MCM, ZAM, and WAM modules enhance segmentation. Table 5 highlights significant improvements in various metrics. Similarly, Fig. 8 clearly shows increased algorithm accuracy, with reduced segmentation errors and more refined details in regions and edges. The specific performance of each configuration is as follows:

(1) UNet Baseline: Serves as the initial framework, establishing fundamental segmentation capabilities. (2) UNet + MCM: Integrating the MCM significantly enhances the recognition of image edges and regional features. MCM optimizes edge perception through morphological operations, improving segmentation precision and structural integrity. As shown in Table 5, there are notable improvements, especially in the HD metric, which decreases from 9.2922 to 8.5472. (3) UNet + MCM + ZAM: Adding the Zernike Attention Module (ZAM) further boosts performance by focusing on key feature areas and enhancing the extraction of deep features, particularly in images with complex background noise or ambiguous features. (4) UNet + MCM + WAM: Incorporating the WAM optimizes feature fusion and improves image reconstruction quality across scales. The multiscale analysis of WAM significantly enhances the ability to capture subtle structural changes. (5) UNet + MCM + ZAM + WAM: The comprehensive application of all modules significantly enhances segmentation detail and precision. ZAM and WAM improve feature discrimination, while MCM ensures structural integrity in the final segmentation results.

Table 3
Metrics Used for Precision Evaluation.

Metric	Formula
Dice Coefficient (DSC) *	$2 \times X \cap Y / (X + Y)$
Intersection over Union (IoU) *	$ X \cap Y / X \cup Y $
Precision (PRE) †	$TP / (TP + FP)$
Recall (REC) †	$TP / (TP + FN)$
Specificity (SPE) †	$TN / (TN + FP)$
Accuracy (ACC) †	$(TP + TN) / (TP + TN + FP + FN)$
Hausdorff Distance (HD) ‡	$\max(h(X, Y), h(Y, X))$

* X (Prediction Region); Y (Ground Truth).

† TP (True Positive); FP (False Positive); TN (True Negative); FN (False Negative).

‡ $h(X, Y)$ (The max distance from a point in X to the closest point in Y); Unit: mm.

Table 4
Performance Comparison of Different Segmentation Methods.

Methods	Dice	IoU	PRE	REC	SPE	ACC	HD	Time(ms)
UNet[7]	0.8378	0.7359	0.8344	0.8570	0.9985	0.9972	9.2922	113.27*
ResUNet[53]	0.8520	0.7614	0.8721	0.8491	0.9989*	0.9975†	8.7410	127.69†
Attention UNet[54]	0.8592	0.7616	0.8770*	0.8622	0.9989*	0.9976*	8.7887	152.12
R2 UNet[55]	0.6141	0.5187	0.7502	0.5678	0.9994	0.9953	11.8522	169.03
Channel UNet[56]	0.8597	0.7632	0.8419	0.8894*	0.9985	0.9975†	8.7437	149.75
KiUNet[57]	0.8460	0.7520	0.8437	0.8675	0.9986	0.9973	8.9189	156.82
TransUNet[9]	0.8269	0.7230	0.8489	0.8297	0.9986	0.9970	9.5646	208.78
MedT[58]	0.8211	0.7171	0.8595	0.8154	0.9987	0.9970	9.6716	389.59
MISSFormer[59]	0.8627†	0.7602	0.8820	0.8739	0.9989*	0.9973	8.6264†	139.25
Swin-UNet[10]	0.8599	0.7589	0.8712	0.8780	0.9987	0.9976*	8.8007	178.63
SAM[60]	0.8604	0.7635†	0.8770*	0.8881†	0.9988†	0.9975†	8.6744	276.24
FB-ZWUNet (Ours)	0.8743*	0.7813*	0.8738†	0.8875	0.9988†	0.9976*	8.2899*	135.72

Precision evaluation methods referenced from Appendix 1.

* indicates the best value, and † indicates the second-best value.

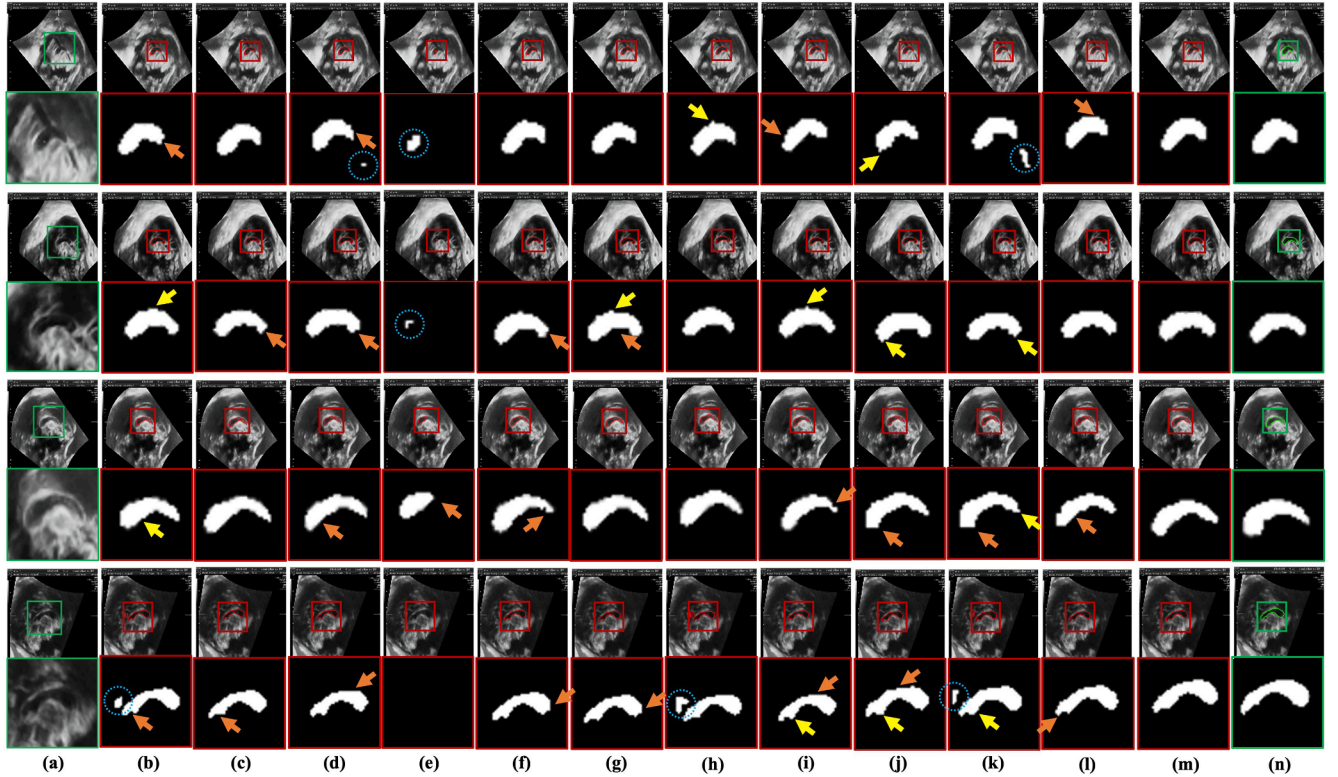


Fig. 6. Qualitative Visual Results of Contrast Experimental. Qualitative visual results on the FB-CC dataset compared with other methods in corpus callosum segmentation task (4 cases). (a) Original images; (b) UNet; (c) Res UNet; (d) Attention UNet; (e) R2 UNet; (f) Channel UNet; (g) KiUNet; (h) TransUNet; (i) MedT; (j) MISSFormer; (k) Swin-Unet; (l) SAM; (m) FB-ZWUNet (Ours); (n) Ground truth. Orange arrow (Missing-segmentation); Yellow arrow (Over-segmentation); Blue circle (Non-contiguous errors). Our method can segment relatively complete CC regions and has a stronger ability to recognize details. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

To demonstrate the contributions of the proposed modules, we visualized and analyzed feature maps before and after introducing key operations (ZAM without Zernike, ZAM with Zernike, WAM without Wavelet, and WAM with Wavelet). As shown in Fig. 9, the following key observations were made: (1) Effect of ZAM: Comparing the encoder output (Before ZAM) with ZAM using a basic attention mechanism (ZAM without Zernike, Fig. 9(b) vs. 9(c)), ZAM improves focus on target regions, enhancing regionalized feature expression and suppressing irrelevant areas. However, boundaries remain unclear due to limited global structural information. (2) Zernike Moments in ZAM: Incorporating Zernike Moments into ZAM (ZAM with Zernike, Fig. 9(c) vs. 9(d)) sharpens boundaries and enhances detail attention, leveraging global feature integration from frequency-domain information to improve local-global feature interaction. (3) Effect of WAM: WAM with a basic

attention mechanism (WAM without Wavelet, Fig. 9(d) vs. 9(e)) enhances multi-scale attention and captures broader contextual information. However, feature integration remains insufficient, and some areas are underrepresented. (4) Wavelet Transformation in WAM: Adding Wavelet Transformation to WAM (WAM with Wavelet, Fig. 9(e) vs. 9(f)) significantly refines feature maps, enhancing fine-grained feature focus, boundary precision, and structural clarity. Wavelet improves multi-scale analysis and suppresses irrelevant regions effectively. ZAM and WAM, combined with Zernike Moments and Wavelet Transformation, improve target region representation, sharpen boundaries, enhance details, and ensure global consistency, enabling high-quality segmentation of the CC.

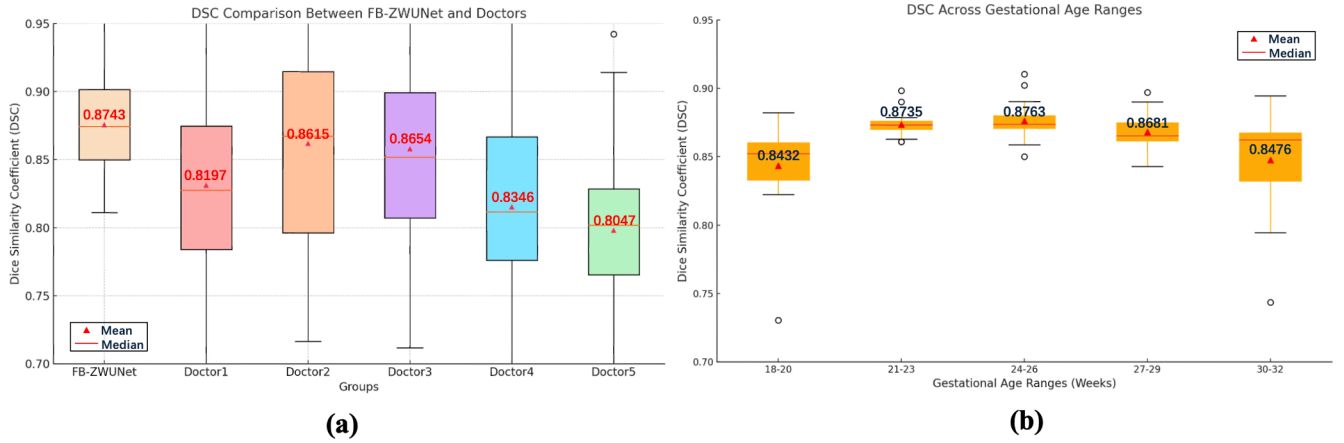


Fig. 7. Practical Perspective Results of Contrast Experimental. (a) DSC Comparison Between FB-ZWUNet and Doctors: This figure shows the comparison of DSC performance between the FB-ZWUNet model and five ultrasound specialists in the task of corpus callosum segmentation. (b) DSC Across Gestational Age Ranges: This figure shows the DSC results of FB-ZWUNet across different gestational age ranges. In both figures, red triangles indicate the mean value for each group, and red horizontal lines represent the median value. The length of the box reflects the interquartile range (IQR), representing segmentation accuracy variation, and the scattered points show the distribution of individual samples within each group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5

Contributions of Various Component Modules in Corpus Callosum Segmentation Task on FB-CC Dataset.

Methods	Dice	IoU	PRE	REC	SPE	ACC	HD
(1) UNet	0.8378	0.7359	0.8344	0.8570	0.9985	0.9972	9.2922
(2) UNet + MCM	0.8406	0.7393	0.8430	0.8532	0.9987	0.9973	8.5472
(3) UNet + MCM + ZAM	0.8553†	0.7565†	0.8765*	0.8680	0.9991	0.9976*	8.4287
(4) UNet + MCM + WAM	0.8514	0.7532	0.8603	0.8781†	0.9991*	0.9975†	8.3840†
(5) UNet + MCM + ZAM + WAM (Ours)	0.8743*	0.7813*	0.8738†	0.8875*	0.9988†	0.9976*	8.2899*

Precision evaluation methods referenced from Appendix 1.

* indicates the best value, and † indicates the second-best value.

3.3.2. Impact of each Loss function

To verify the effectiveness of the composite loss functions introduced by MCM in supervising edge and regional information of CC, an ablation study on loss functions was conducted. In the experiments, Dice loss served as the basis for region-based loss and BCE loss for edge-based loss, with additional loss functions evaluated for their impacts. The results in Table 6 indicate that after incorporating Focal loss, performance improves from 84.52 % to 84.91 %, and similarly, with Tversky loss, it increases to 85.17 %. Furthermore, the Hausdorff distance decreases from 8.5179 to 8.3677 and 8.3245, respectively. The application of the final composite loss function leads to a reduction of 0.228 in HD compared to the baseline loss function, highlighting the superior performance of the mixed loss approach in supervising edge and region predictions within MCM.

The preconditioning loss functions, integral to MCM, not only focus on morphological features of CC, such as contours and regions but also integrate various loss functions to enhance segmentation performance. As depicted in Fig. 10, MCM enhances the accuracy of overall region segmentation through Dice loss; BCE loss concentrates on pixel-level classification, essential for capturing detailed edges and contours. Focal loss addresses class imbalance in classification problems by reducing focus on easily classified examples and increasing attention to difficult or misclassified examples, thereby improving segmentation of challenging areas. Tversky loss, by adjusting weights for false positives and false negatives, further optimizes handling of imbalanced data. The combination of these loss functions not only increases the precision of CC segmentation but also ensures robustness and high performance of the model amidst morphological variations. With the addition of these loss functions, the depiction of contours and details becomes more accurate.

3.3.3. Impact of training Hyper-Parameters

In this part, we conduct a grid search for the hyper-parameters α_1 and α_2 in the losses L_{Region} and L_{Edge} during the training process. We fix the hyper-parameters β_1 and β_2 to 1 because β_1 and β_2 control the weights of Dice Loss and BCE Loss, respectively. These loss functions have been widely validated to be effective in handling class imbalance and boundary accuracy. By fixing these hyper-parameters, we can focus on verifying the impact of α_1 and α_2 on the loss function.

We use a grid search to simultaneously vary α_1 and α_2 , covering all reasonable combinations. The values of α_1 and α_2 are set to [0.2, 0.4, 0.6, 0.8, 1]. We evaluate every combination of α_1 and α_2 . As shown in Fig. 11, the segmentation accuracy increases with the increase of α_1 and decreases with the increase of α_2 . Therefore, we set the hyper-parameter combination $(\alpha_1, \beta_1, \alpha_2, \beta_2)$ to (1, 1, 0.2, 1).

3.3.4. Impact of edge operator in MCM

To assess the robustness of our model in handling inaccuracies in edge marking for corpus callosum segmentation, we conducted ablation experiments on the FB-CC dataset. These experiments compared the effects of different edge operators on generating edge ground truths for training. As shown in Fig. 12, the performance using Sobel [61], Roberts [62], Prewitt [63], Scharr [64], and Laplacian [65] operators was comparable to that of the Canny [47] operator, even with slight inaccuracies in edge marking. This demonstrates that our model is highly robust to minor variations in boundary ground truths generated by different edge operators. Additionally, the Canny operator performed the best among these operators, so we chose Canny as the primary image processing method.

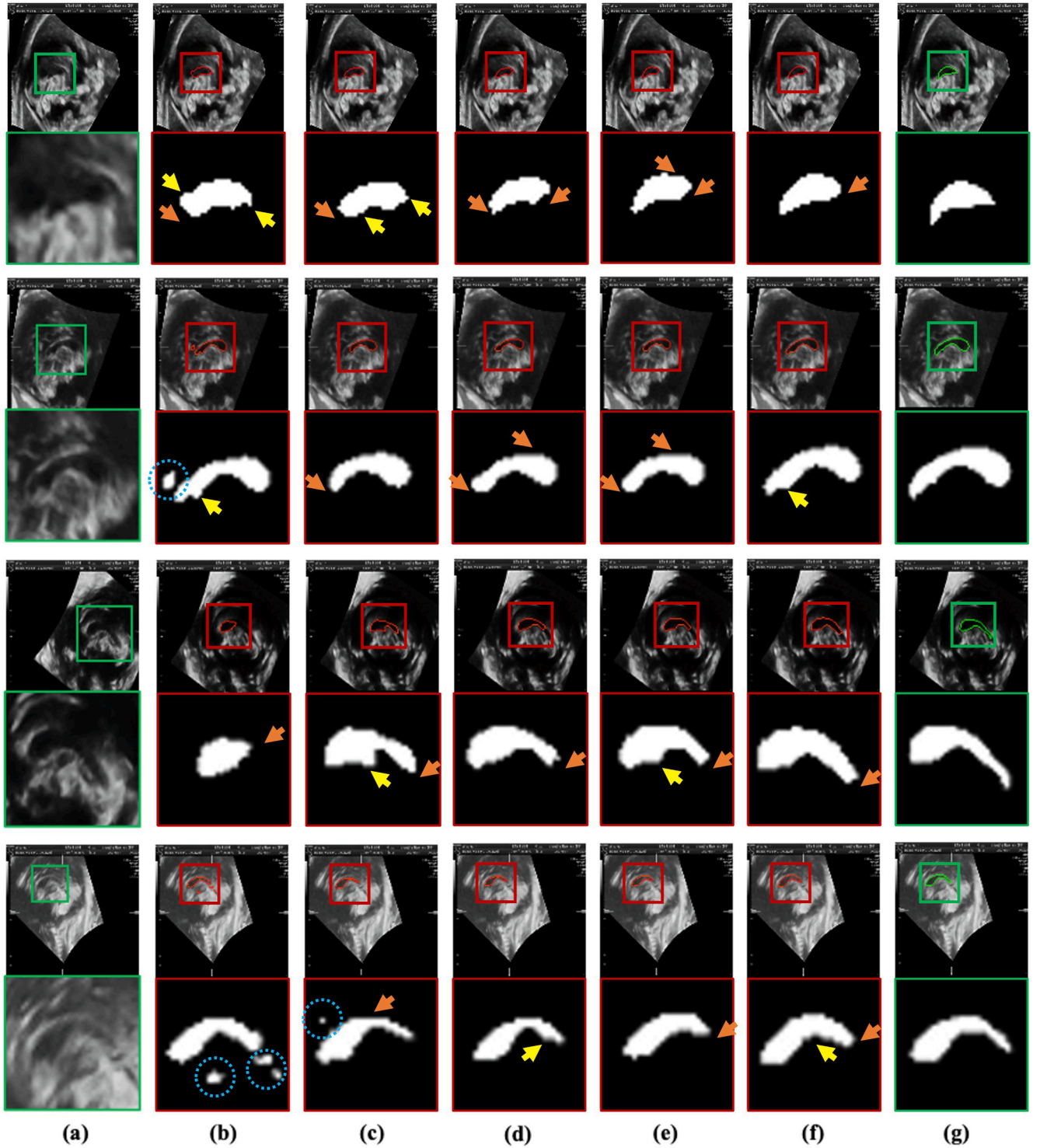


Fig. 8. Module Composition Ablation Experiment Results. Visual comparison of corpus callosum segmentation results among different configurations (4 cases): (a) Original images; (b) UNet; (c) UNet + MCM; (d) UNet + MCM + ZAM; (e) UNet + MCM + WAM; (f) UNet + MCM + ZAM + WAM (Ours); (g) Ground truth. Orange arrow (Missing-segmentation); Yellow arrow (Over-segmentation); Blue circle (Non-contiguous errors). Note that segmentation performance improves with the addition of each component. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Discussion and Conclusion

4.1. Performance and Practical applications

In this study, we introduced FB-ZWUNet, an end-to-end network that achieved a DSC of 0.8743 and an IoU of 0.7813 for automated CC segmentation in fetal brain ultrasound images, outperforming existing

state-of-the-art models. By integrating the Zernike Attention Module (ZAM), Wavelet Attention Module (WAM), and Morphological Constraint Module (MCM), the model effectively enhanced feature extraction, fusion, and edge detection. Specifically, ZAM emphasized shape and texture characteristics, WAM improved the multi-scale fusion of features, and MCM ensured precise delineation of edges and regional details. The synergy of these three modules significantly improved

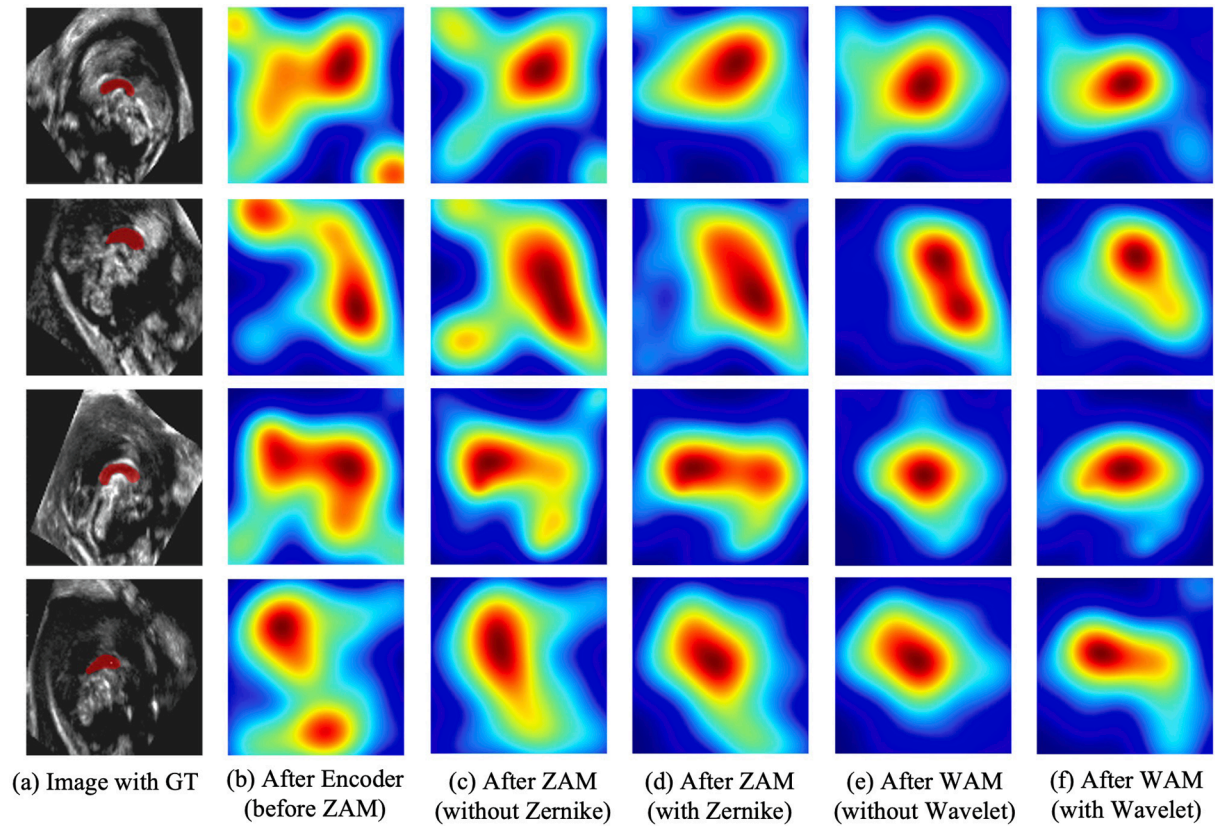


Fig. 9. Visualization of feature maps from ZAM and WAM modules. From left to right are (a) original image with GT, (b) output from encoder (before ZAM), (c) ZAM with basic attention mechanism (without Zernike), (d) ZAM with Zernike Moments (with Zernike), (e) WAM with basic attention mechanism (without Wavelet), and (f) WAM with Wavelet Transformation (with Wavelet), respectively.

Table 6

Contributions of Various Loss Functions in Corpus Callosum Segmentation Task on FB-CC Dataset.

<i>Loss Region</i>		<i>Loss Edge</i>		<i>Dice</i>	<i>IoU</i>	<i>PRE</i>	<i>REC</i>	<i>SPE</i>	<i>ACC</i>	<i>HD</i>
<i>Dice Loss</i>	<i>Focal Loss</i>	<i>BCE Loss</i>	<i>Tversky Loss</i>							
✓		✓		0.8452	0.7447	0.8628†	0.8324	0.9831	0.9972	8.5179
✓	✓	✓		0.8491	0.7534	0.8412	0.8752†	0.9982	0.9974†	8.3677
✓		✓	✓	0.8517†	0.7557†	0.8552	0.8649	0.9985†	0.9973	8.3245†
✓	✓	✓	✓	0.8743*	0.7813*	0.8738*	0.8875*	0.9988*	0.9976*	8.2899*

Precision evaluation methods referenced from Appendix 1.

* indicates the best value, and † indicates the second-best value.

segmentation accuracy and robustness.

Comparisons with experienced ultrasonographers revealed that while manual annotations are generally accurate, they may vary due to subjective factors such as individual experience. FB-ZWUNet not only matched but also surpassed the consistency achieved by expert manual annotations, thereby reducing reliance on specialist expertise and minimizing human errors. This suggests that automated approaches can offer more stable and reproducible results, ultimately contributing to increased reliability and efficiency in clinical practice.

Nonetheless, certain limitations remain. Ultrasound images affected by strong cranial shadowing pose challenges for both clinicians and the algorithm, as acoustic shadows obscure critical CC boundaries. In such cases, even manual annotations may rely on estimations, introducing subjectivity. Additionally, although our dataset covers various gestational ages, devices, and participants, further expansion is needed to improve generalizability. Incorporating data from more diverse geographic regions, clinical settings, and imaging conditions can bolster the adaptability to complex, real-world scenarios.

Despite these challenges, the successful application of FB-ZWUNet to

real clinical data underscores its practicality. As shown in Fig. 13, we integrated the model into a comprehensive computer-aided diagnosis (CAD) system that supports real-time prenatal evaluations. The system accommodates flexible data inputs—either directly from ultrasound machines or via cameras on PCs and mobile devices—and includes preprocessing modules to correct geometric distortions, enhance image clarity, and calculate scales for accurate CC measurements. With these optimizations, FB-ZWUNet consistently delivers high-precision segmentation results that are seamlessly integrated into diagnostic reports. These reports include segmentation overlays, measurements, and clinical interpretations, enabling clinicians to make informed decisions quickly. The model's efficient processing meets real-time clinical requirements, streamlining the diagnostic process and improving overall prenatal care.

4.2. Generalization and future work

While FB-ZWUNet has demonstrated strong performance, opportunities for further improvement remain. Ultrasound image quality can

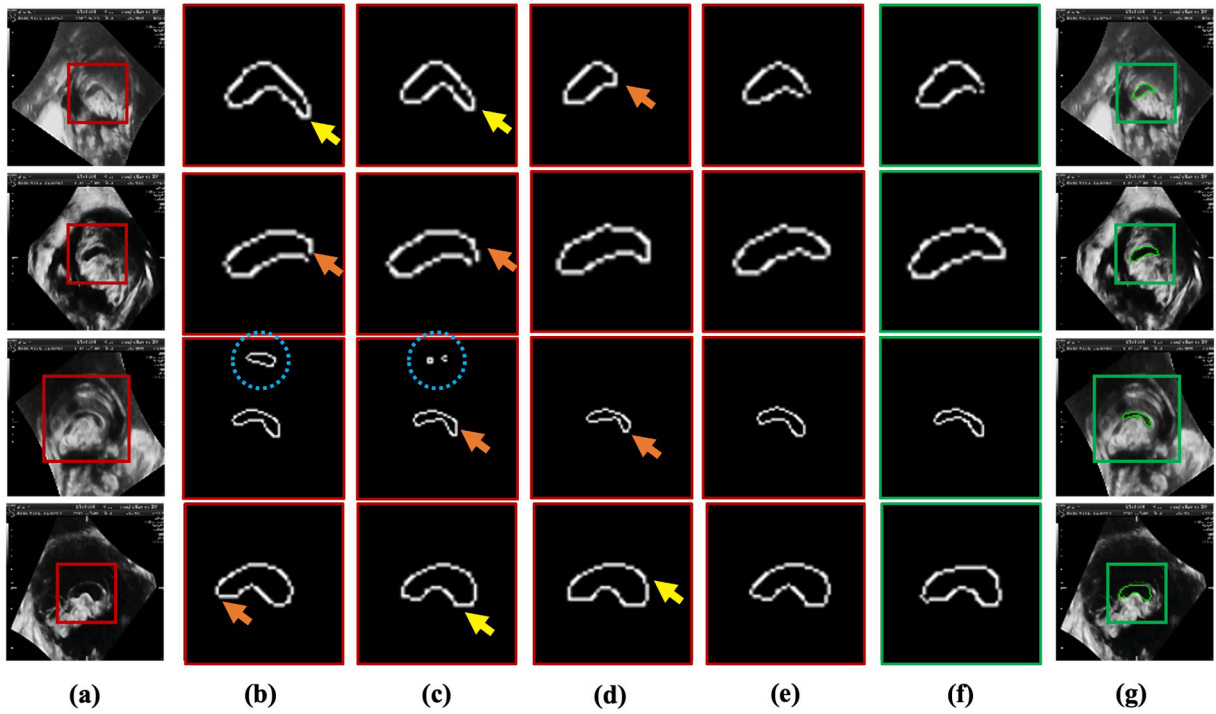


Fig. 10. Loss Function Ablation Experiment Results. Edge prediction in the MCM edge branch compared with different loss functions (4 cases): (a) Original images; (b) Dice Loss + BCE Loss; (c) Dice Loss + Focal Loss + BCE Loss; (d) Dice Loss + BCE Loss + Tversky Loss; (e) Dice Loss + Focal Loss + BCE Loss + Tversky Loss (Ours); (f) Ground truth – edge; (g) Ground truth – mask. Orange arrow (Missing-segmentation); Yellow arrow (Over-segmentation); Blue circle (Non-contiguous errors). Our combination of loss functions better constrains the model to complete edge prediction, providing supplementary boundary information for corpus callosum segmentation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

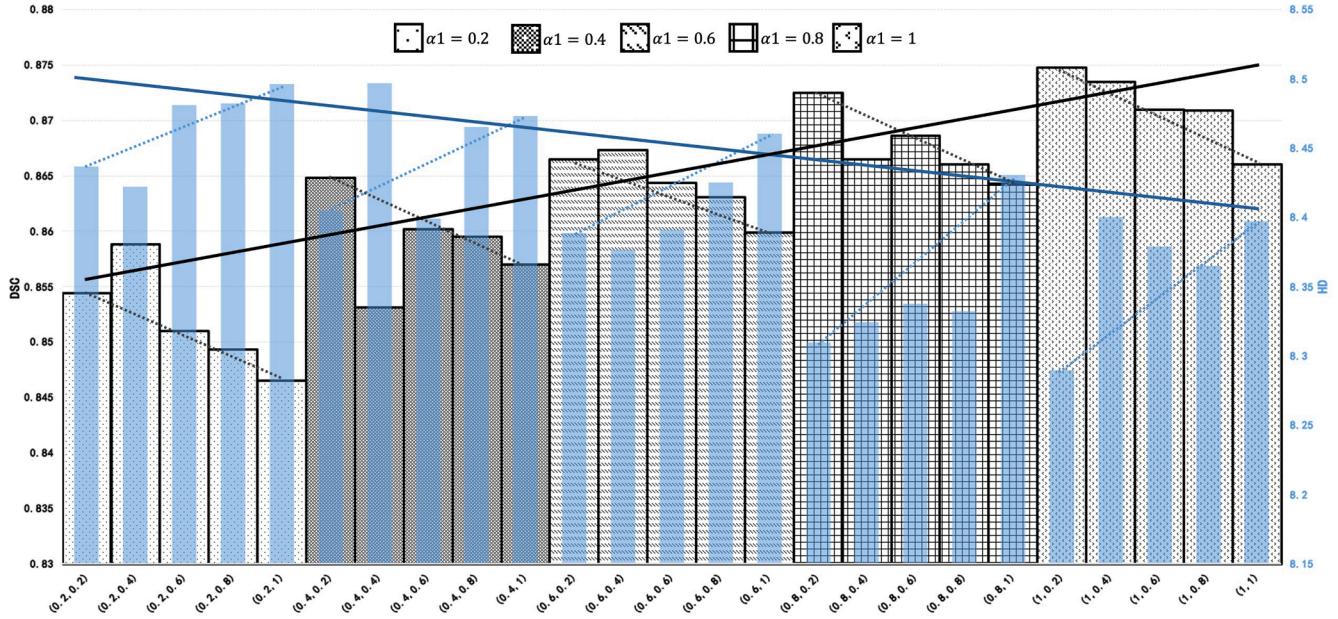


Fig. 11. Training Hyper-parameters Ablation Experiment Results. The black bar chart represents the Dice coefficient (DSC) corresponding to different α_1 and α_2 settings, with β_1 and β_2 fixed at 1. The blue bar chart shows the corresponding Hausdorff Distance (HD). The solid line indicates the trend of increasing segmentation accuracy with increasing α_1 , while the dashed line shows the trend of decreasing segmentation accuracy with increasing α_2 . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

vary substantially due to differences in equipment, operator skill, and patient characteristics. Although our current dataset includes multiple devices, extending the dataset's scope is essential. Incorporating data from a broader range of ultrasound equipment, geographic regions, and medical institutions will help the model adapt to diverse imaging

conditions and patient populations, thus enhancing its robustness.

In parallel, exploring advanced deep learning methodologies can further improve segmentation accuracy and efficiency. Transformer-based architectures, which excel at modeling long-range dependencies, may offer new avenues for refining image feature

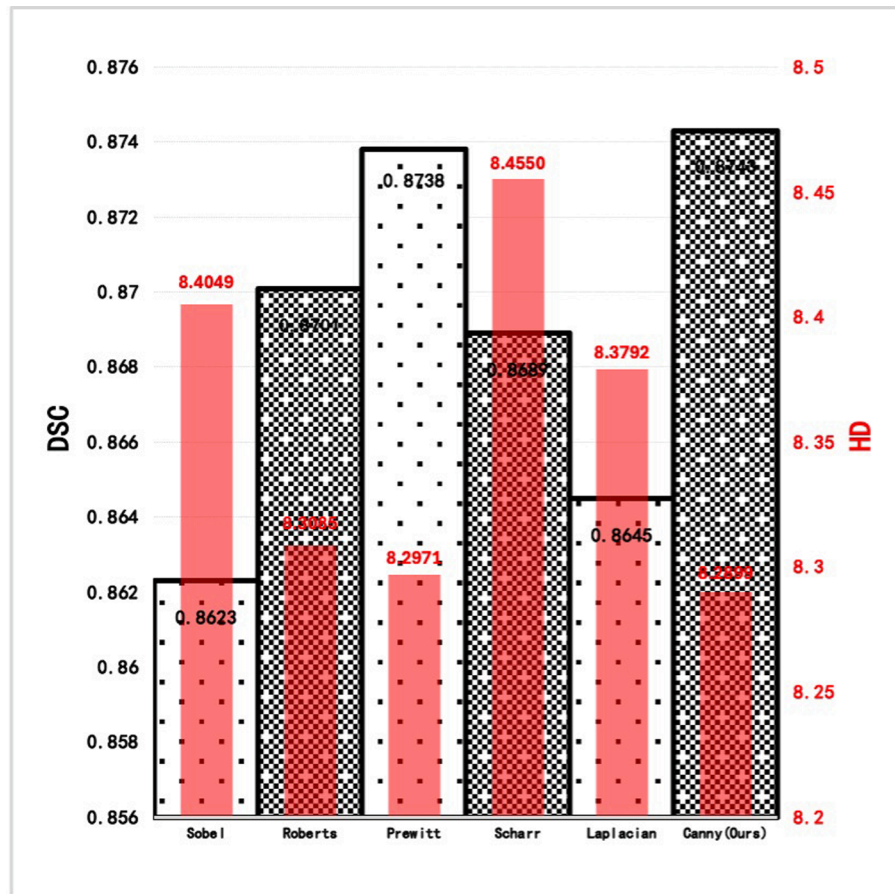


Fig. 12. Edge Operator Ablation Experiment Results. Impact of edge operator in the MCM experimental results. The black bar chart represents the Dice coefficient (DSC) of the network when different edge operators are applied for image processing, and the red bar chart shows the corresponding Hausdorff Distance (HD). The DSC for all operators exceeds 0.86, and the HD are all below 8.5. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

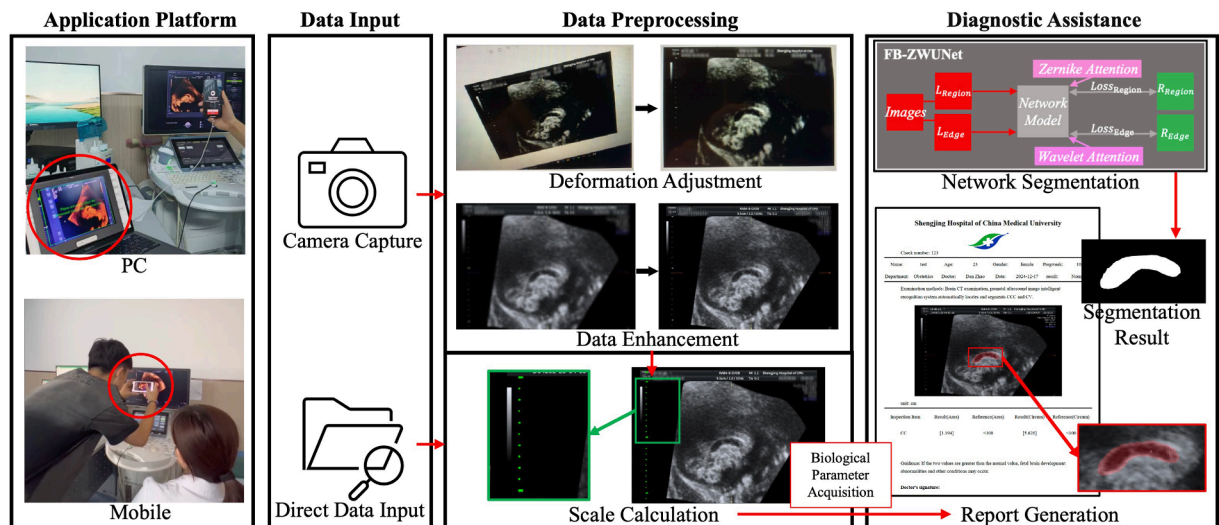


Fig. 13. Overview of the FB-ZWUNet Integrated CAD System Workflow. The CAD system supports both PC and mobile platforms, providing flexible and accessible tools for clinical use. The workflow includes three main stages: Data Input: Ultrasound images are captured directly from ultrasound devices or external cameras (PC/Mobile). Data Preprocessing: Includes deformation adjustment, data enhancement for improved clarity, and scale calculation for accurate biological measurements. Diagnostic Assistance: FB-ZWUNet performs automatic segmentation of the corpus callosum (CC), generating precise segmentation results. This information is used to produce comprehensive diagnostic reports, aiding clinicians in prenatal diagnostics.

representations. Likewise, self-supervised learning techniques can harness unlabeled data to strengthen model generalization. By integrating such approaches, FB-ZWUNet can become even more adept at handling complex clinical scenarios and evolving imaging technologies.

Another key area for future work involves increasing the quantity and quality of annotated data, including comprehensive measurements of the CC (e.g., perimeter, area, thickness). Such data would enable the establishment of reliable reference standards for different developmental stages, supporting more nuanced clinical assessments. Rigorous large-scale validations are needed to fully evaluate the model's impact on diagnostic accuracy, efficiency, and clinical decision-making processes. By continuously expanding the dataset, embracing advanced learning techniques, and refining evaluation criteria, FB-ZWUNet can evolve into a more robust, accurate, and clinically valuable tool, ultimately contributing to improved prenatal diagnostics and neonatal health outcomes.

4.3. Conclusion

FB-ZWUNet represents a noteworthy advancement in automated fetal brain ultrasound segmentation, blending an innovative network architecture with a well-annotated dataset. By effectively capturing the morphological and textural characteristics of the CC, the model ensures accurate and reliable segmentation. Its integration into a mobile CAD system demonstrates tangible clinical value, providing real-time assistance in prenatal diagnostics and reducing dependence on specialist expertise. Addressing the current limitations and pursuing the proposed improvements will further enhance the model's adaptability and accuracy. Ultimately, FB-ZWUNet can facilitate earlier detection and intervention for fetal neurological conditions, promoting better prenatal care and improving neonatal health outcomes.

CRedit authorship contribution statement

Qifeng Wang: Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Dan Zhao:** Resources, Data curation. **Hao Ma:** Writing – review & editing, Validation, Software. **Bin Liu:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the National Natural Science Foundation of China (Nos. 62372079, 61972440 and 61572101), the Fundamental Research Funds for the Central Universities of China (No. DUT24YG129), the Natural Science Foundation of Liaoning Province of China (No. 2021-YGJC-23 and 2022-YGJC-73).

Footnote.

Ethical Statement: All procedures carried out in studies involving human participants comply with the ethical standards of institutions and/or national research committees, as well as the 1964 Declaration of Helsinki and its subsequent amendments or similar ethical standards. All individuals involved in the study obtained informed consent. This study was approved by the Ethics Committee of Shengjing Hospital Affiliated to China Medical University with the ethics number 2023PS910K.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bspc.2025.107499>.

Data availability

Data will be made available on request.

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